

Customer Shopping Behavior Analysis

Project Overview:

This project analyzes customer shopping behavior using transactional data from 3,900 purchases across various product categories. The goal is to uncover insights into spending patterns, customer segments, product preferences, and subscription behavior to guide strategic business decisions.

Dataset Summary:

The dataset had the following details, it include rows, columns and various other diversification that are required.

Rows: 3,900

Columns: 18

Key Features:

- **Customer demographics** (Age, Gender, Location, Subscription Status)
- **Purchase details** (Item Purchased, Category, Purchase Amount, Season, Size, Color)
- **Shopping behavior** (Discount Applied, Promo Code Used, Previous Purchases, Frequency of Purchases, Review Rating, Shipping Type)

Missing Data: 37 values in Review Rating column

Exploratory Data Analysis using Python:

We began with data preparation and cleaning in Python:

- **Data Loading:** Imported the dataset using pandas.
- **Initial Exploration:** Used

`df.info()`

to check structure and

`.describe()`

for summary statistics.

[8]:

	Customer ID	Age	Gender	Item Purchased	Category	Purchase Amount (USD)	Location	Size	Color	Season	Review Rating	Subscription Status	Shipping Type	Discount Applied	Promo Code Used	Previous Purchases	Payment Method
0	1	55	Male	Blouse	Clothing	53	Kentucky	L	Gray	Winter	3.1	Yes	Express	Yes	Yes	14	Venmo
1	2	19	Male	Sweater	Clothing	64	Maine	L	Maroon	Winter	3.1	Yes	Express	Yes	Yes	2	Cash
2	3	50	Male	Jeans	Clothing	73	Massachusetts	S	Maroon	Spring	3.1	Yes	Free Shipping	Yes	Yes	23	Credit Card
3	4	21	Male	Sandals	Footwear	90	Rhode Island	M	Maroon	Spring	3.5	Yes	Next Day Air	Yes	Yes	49	PayPal
4	5	45	Male	Blouse	Clothing	49	Oregon	M	Turquoise	Spring	2.7	Yes	Free Shipping	Yes	Yes	31	PayPal

	Shipping Type	Discount Applied	Promo Code Used	Previous Purchases	Payment Method	Frequency of Purchases
5	Express	Yes	Yes	14	Venmo	Fortnightly
5	Express	Yes	Yes	2	Cash	Fortnightly
5	Free Shipping	Yes	Yes	23	Credit Card	Weekly
5	Next Day Air	Yes	Yes	49	PayPal	Weekly
5	Free Shipping	Yes	Yes	31	PayPal	Annually

- **Missing Data Handling:** Checked for null values and imputed missing values in the Review Rating column using the median rating of each product category.
- **Column Standardization:** Renamed columns to snake case for better readability and documentation.
- **Feature Engineering:**
 - Created **age_group** column by binning customer ages.
 - Created **purchase_frequency_days** column from purchase data.
- **Data Consistency Check:** Verified if **discount_applied** and **promo_code_used** were redundant;
dropped promo_code_used

- **Database Integration:** Connected Python script to PostgreSQL and loaded the cleaned DataFrame into the database for SQL analysis.

```
!pip install pandas sqlalchemy psycopg2-binary
```

```
Requirement already satisfied: pandas in d:\python\lib\site-packages (2.2.2)
Requirement already satisfied: sqlalchemy in d:\python\lib\site-packages (2.0.34)
Requirement already satisfied: psycopg2-binary in d:\python\lib\site-packages (2.9.12)
Requirement already satisfied: numpy>=1.26.0 in d:\python\lib\site-packages (from pandas) (1.26.4)
Requirement already satisfied: python-dateutil>=2.8.2 in d:\python\lib\site-packages (from pandas) (2.9.0.post0)
Requirement already satisfied: pytz>=2020.1 in d:\python\lib\site-packages (from pandas) (2024.1)
Requirement already satisfied: tzdata>=2022.7 in d:\python\lib\site-packages (from pandas) (2023.3)
Requirement already satisfied: typing-extensions>=4.6.0 in d:\python\lib\site-packages (from sqlalchemy) (4.11.0)
Requirement already satisfied: greenlet!=0.4.17 in d:\python\lib\site-packages (from sqlalchemy) (3.0.1)
Requirement already satisfied: six>=1.5 in d:\python\lib\site-packages (from python-dateutil>=2.8.2->pandas) (1.16.0)
```

```
[62]: import pandas as pd
      from sqlalchemy import create_engine

      # 2. Define your PostgreSQL connection credentials
      username = "postgres"
      password = "ishi1606"           # Your PostgreSQL password
      host = "localhost"              # Running on your Local machine
      port = "5432"                   # Default PostgreSQL port
      database = "customer_behaviour" # The exact database you just created

      # 3. Create the database connection engine
      engine = create_engine(f"postgresql+psycopg2://{username}:{password}@{host}:{port}/{database}")

      # 4. Choose a name for your table in pgAdmin
      table_name = "customer_behaviour"

      # 5. Push the DataFrame to pgAdmin 4
      # 'if_exists="replace"' will create the table if it doesn't exist,
      # or overwrite it if it already exists.
      df.to_sql(table_name, engine, if_exists="replace", index=False)

      print(f"🎉 Success! Data successfully loaded into table '{table_name}' inside database '{database}'.")
      🎉 Success! Data successfully loaded into table 'customer_behaviour' inside database 'customer_behaviour'.
```

Activate Windows

Data Analysis using SQL (Business Transactions):

Some of the queries performed to answer key business question:

Ques1) Revenue by Gender – Compared total revenue generated by male vs. female customers.

Input:

```
--- what is the total revenue generated by male vs female customers?
select gender, sum(purchase_amount) as revenue
from customer_behaviour
group by gender;
```

Output:

Data Output		Messages	Notifications
			
			
			

Ques2) High-Spending Discount Users – Identified customers who used discounts but still spent above the average purchase amount.

Input:

```
-- which customer used a discount but still spent more than average purchase amount?
SELECT customer_id, purchase_amount
FROM customer_behaviour
WHERE discount_applied = 'Yes'
    AND purchase_amount >= (SELECT AVG(purchase_amount) FROM customer_behaviour);
```

Output:

	customer_id bigint	purchase_amount bigint
1	2	64
2	3	73
3	4	90
4	7	85
5	9	97
6	12	68
7	13	72
8	16	81
9	20	90
10	22	62
11	24	88
12	29	94
13	32	79
14	33	67
15	35	91
16	37	60

Total rows: 37 Query complete 00:00:00.186

Ques3) Top 5 Products by Rating – Found products with the highest average review ratings.

Input:

```
--- which are the top 5 products with the highest avergae review rating?
select item_purchased , round(avg(review_rating::numeric),2) as "average product rating"
from customer_behaviour
group by item_purchased
order by avg(review_rating) desc
limit(5);
```

Output:

Data Output		Messages	Notifications
	item_purchased text	average product rating numeric	
1	Gloves	3.86	
2	Sandals	3.84	
3	Boots	3.82	
4	Hat	3.80	
5	Skirt	3.78	
Total rows: 5		Query complete 00:00:00.143	

Ques4) Shipping Type Comparison – Compared average purchase amounts between Standard and Express shipping.

Input:

```
--- compare the average purchase amounts between standard and express shipping.
select shipping_type, round(avg(purchase_amount),2)
from customer_behaviour
where shipping_type in ('Standard','Express')
group by shipping_type;
```

Output:

Data Output			Messages	Notifications
	shipping_type	round		
	text	numeric		
1	Standard	58.46		
2	Express	60.48		

Ques5) Subscribers vs. Non-Subscribers – Compared average spend and total revenue across subscription status.

Input:

```
Query Query History
25
26 --- do subscribed customers spend more? compare average spend and total revenue between subscribers
27 ---and non subscribers.
28 select subscription_status,
29 count(customer_id) as total_customer,
30 Round(avg(purchase_amount),2) as average_spend,
31 Round(sum(purchase_amount),2) as total_revenue
32 from customer_behaviour
33 group by subscription_status
34 order by average_spend,total_revenue desc;
```

Output:

Data Output			Messages	Notifications
	subscription_status	total_customer	average_spend	total_revenue
	text	bigint	numeric	numeric
1	Yes	1053	59.49	62645.00
2	No	2847	59.87	170436.00

Ques6) Discount-Dependent Products – Identified 5 products with the highest percentage of discounted purchases.

Input:

```
--- which 5 products have the highest percentage of purchases with discount applied?
SELECT item_purchased,
       ROUND(100 * SUM(CASE WHEN LOWER(discount_applied) = 'yes' THEN 1 ELSE 0 end)/ COUNT(*), 2) AS discount_rate
FROM customer_behaviour
GROUP BY item_purchased
ORDER BY discount_rate DESC
LIMIT 5;
```

Output:

Data Output			Messages	Notifications
	item_purchased text	discount_rate numeric		
1	Hat	50.00		
2	Sneakers	49.00		
3	Coat	49.00		
4	Sweater	48.00		
5	Pants	47.00		

Ques7) Customer Segmentation – Classified customers into New, Returning, and Loyal segments based on purchase history.

Input:

```
--- segment customers into new, returning and loyal based on the total number of previous
--- purchases and show the count of each segment.
with customer_type as (
select customer_id, previous_purchases,
case
when previous_purchases = 1 then 'New'
when previous_purchases between 2 and 10 then 'Returning'
else 'Loyal'
end as customer_segment
from customer_behaviour
)
select customer_segment, count(*) as "Number Of Customers"
from customer_type
group by customer_segment;
```

Output:

Data Output		Messages	Notifications
	customer_segment text	Number Of Customers bigint	
1	Loyal	3116	
2	New	83	
3	Returning	701	

Ques8) Top 3 Products per Category – Listed the most purchased products within each category.

Input:

```
58
59 --- what are the top 3 most purchased products within each category.
60 with item_counts as (
61   select category, item_purchased,
62   count(customer_id) as total_orders,
63   row_number() over (partition by category order by Count(customer_id)desc) as item_rank
64   from customer_behaviour
65   group by category, item_purchased
66 )
67 select item_rank, category, item_purchased, total_orders
68 from item_counts
69 where item_rank <= 3;
```

Output:

Data Output

Messages

Notifications

item_rank
bigint

category
text

item_purchased
text

totalOrders
bigint

1

2

3

4

5

6

7

8

9

10

11

1

2

3

1

2

3

1

2

3

1

2

Accessori...

Accessori...

Accessori...

Clothing

Clothing

Clothing

Footwear

Footwear

Footwear

Outerwear

Outerwear

Jewelry

Sunglasses

Belt

Blouse

Pants

Shirt

Sandals

Shoes

Sneakers

Jacket

Coat

171

161

161

171

171

169

160

150

145

163

161

Ques9) Repeat Buyers & Subscriptions – Checked whether customers with >5 purchases are more likely to subscribe.

Input:

```
--- are customers who are repeat buyers (more than 5 previous_purchases) also likely to subscribe?
select subscription_status,
count(customer_id) as repeat_buyers
from customer_behaviour
where previous_purchases > 5
group by subscription_status;
```

Output:

Data Output	Messages	Notifications
		
	subscription_status text	repeat_buyers bigint
1	No	2518
2	Yes	958

Ques10) Revenue by Age Group – Calculated total revenue contribution of each age group.

Input:

```
--- what is the revenue contribution of each age group?
select age_group,
sum(purchase_amount) as total_revenue
from customer_behaviour
group by age_group
order by total_revenue desc;
```

Output:

	age_group text	total_revenue numeric
1	young adult	62143
2	middle-aged	59197
3	adult	55978
4	senior	55763

Dashboard in Power BI:

At last, I've built an interactive dashboard in Power BI to present insights visually.



Business Recommendations:

- Boost Subscriptions – Promote exclusive benefits for subscribers.
- Customer Loyalty Programs – Reward repeat buyers to move them into the “Loyal” segment.
- Review Discount Policy – Balance sales boosts with margin control.
- Product Positioning – Highlight top-rated and best-selling products in campaigns.
- Targeted Marketing – Focus efforts on high-revenue age groups and express-shipping users.

